

CHANDRASHEKAR BALA

Cybersecurity Researcher | Security Engineering | VAPT

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SUMMARY

Cybersecurity professional and Team Lead focused on security assessment, attack-path analysis, evidence-driven investigation, and practical security engineering. Experience spans authorized web application assessment, Vulnerability Assessment and Penetration Testing (VAPT), network and wireless security, threat intelligence, malware and digital forensics research, exploit analysis, and Linux systems engineering. Combines attacker-informed testing with manual validation, technical documentation, impact analysis, and remediation-focused recommendations. Develops security tooling and investigates systems-level issues using Python, Bash, C, Linux, and network-analysis tools.

EXPERIENCE

Team Lead – Cybersecurity, NETWORKWALKS Technologies Oct 2026 – Present
Security Assessment | Offensive and Defensive Security | Adversary Research | Security Engineering

- Lead and contribute to cybersecurity project execution across assessment, investigation, adversary research, and technical work.
- Guide reconnaissance, vulnerability validation, attack-path analysis, evidence review, impact assessment, and technical reporting.
- Coordinate assigned activities and translate validated findings into practical remediation, hardening, detection, and engineering recommendations.

Cybersecurity Intern, NETWORKWALKS Technologies Sep 2026 – Oct 2026
VAPT | Network Security | Security Operations | Digital Forensics

- Performed reconnaissance, service enumeration, vulnerability assessment, manual validation, evidence review, and security documentation.
- Applied offensive and investigative techniques to analyze weaknesses, security events, attack paths, technical impact, and remediation options.

Cybersecurity Researcher – Independent Jul 2024 – Present
Independent Cybersecurity Research and Security Engineering

- Conduct practical research across web, network, wireless, application, and systems security, including vulnerability validation, controlled exploitation, privilege-escalation research, and attack-path analysis in authorized or isolated environments.
- Analyze HTTP requests, authentication and access-control behavior, network traffic, OSINT sources, threat reports, indicators, and adversary tactics, techniques, and procedures (TTPs), using MITRE ATT&CK to structure adversary analysis.
- Perform controlled malware and Windows-forensics research involving suspicious files, registry artifacts, filesystem and disk evidence, execution behavior, and static/dynamic analysis.
- Develop Python and Bash utilities for repeatable security analysis and research workflows; document evidence, technical impact, and remediation considerations.

FEATURED SECURITY WORK

Mediroza General Hospital – Authorized Black-Box Web Application Security Assessment Sep – Oct 2026
Web Application Security | Burp Suite | Nmap | Manual Validation | CWE/CVSS

- Reconstructed an attack path from reconnaissance through demonstrated SQL injection and authentication bypass to protected report retrieval and impact analysis.
- Documented eight findings involving directory exposure, internal database-backup exposure, SQL injection, verbose errors, sensitive report retrieval, predictable storage paths, error-log disclosure, and technology fingerprinting.
- Produced evidence-led findings covering exploitation conditions, root cause, impact, severity context, remediation, and retest considerations. Restricted records and sensitive assessment evidence are excluded from public materials.

Case record: GitHub assessment repository

RTL8812BU / RTL8822BU Linux Driver Modernization 2026
C | Linux Kernel | GCC | Git | Real-Hardware Validation

- Analyzed kernel-facing C code and implemented targeted compatibility changes across 13 source files for newer Linux kernel APIs and build requirements.
- Diagnosed compiler, kernel API, and module-build issues using kernel headers, GCC, conditional compilation, Git, and kernel build workflows.
- Built and validated the modified module on RTL8812BU-based hardware; investigated wireless testing workflows involving monitor mode, packet capture, Wireshark, and Aircrack-ng.

Repository: RTL8812BU Linux driver project

ADDITIONAL SECURITY PROJECTS

Network Reconnaissance and Attack-Surface Analysis

Security Research

Nmap/NSE | WHOIS | DNS | Certificate Discovery | TLS | OSINT

- Conducted structured reconnaissance using DNS and WHOIS tooling, certificate discovery, web fingerprinting, TLS inspection, and service/version enumeration.
- Preserved scan artifacts and correlated host, service, web, and infrastructure observations, distinguishing confirmed evidence from hypotheses.

Web Application Security Research

Feb 2026 – Present

PortSwigger Web Security Academy | Burp Suite | OWASP ZAP

- Completed 50+ PortSwigger Web Security Academy labs covering SQL injection, authentication, access control, XSS, session behavior, and HTTP request/response analysis.
- Reproduced lab vulnerabilities and examined exploitation conditions, security impact, and remediation approaches.

Vulnerability Assessment and Privilege-Escalation Research

Security Research

VulnHub | Nmap | Metasploit | Linux

- Perform reconnaissance, service enumeration, vulnerability validation, controlled exploitation, privilege-escalation research, and post-exploitation analysis in isolated virtual environments.
- Trace attack paths and document evidence, exploitation conditions, impact, and hardening recommendations.

Buffer-Overflow and Exploit Research

Security Research

C | Linux | GDB | Memory and Control-Flow Analysis

- Investigate stack-based buffer-overflow behavior through stack layout, register inspection, memory analysis, debugging, and controlled proof-of-concept workflows.

Malware Analysis and Windows Forensics Research

Security Research

Ghidra | Volatility | Wireshark | Static/Dynamic Analysis

- Examine suspicious files, binary behavior, registry artifacts, filesystem and disk evidence, indicators, and host/network evidence in controlled research exercises.

Security Automation and Proof-of-Concept Tooling

Independent Research

Python | Bash | Linux

- Develop scripts and utilities for reconnaissance, repeatable analysis, security testing, and research automation.

TECHNICAL SKILLS

VAPT and Offensive Security	Reconnaissance, service enumeration, vulnerability assessment and validation, controlled exploitation, privilege escalation, post-exploitation analysis, attack-path reconstruction
Web and Application Security	OWASP Top 10, SQL injection, cross-site scripting (XSS), authentication, authorization, access control, session behavior, HTTP/HTTPS analysis, Burp Suite, OWASP ZAP
Network and Wireless Security	TCP/IP, DNS, DHCP, routing, subnetting, Nmap/NSE, packet capture, traffic analysis, Aircrack-ng, wireless testing workflows
Defensive Security	Security monitoring, log and alert analysis, indicators of compromise (IOCs), network evidence correlation, incident-analysis workflows, remediation and detection considerations
Threat Intelligence	OSINT, threat-report analysis, infrastructure analysis, domains/IPs, indicator analysis, adversary behavior, TTP analysis, MITRE ATT&CK
Malware and Forensics	Static/dynamic analysis, suspicious-file investigation, Windows registry artifacts, filesystem and disk evidence, memory and execution analysis, Ghidra, Volatility
Exploit and Systems Research	C, Linux, GDB, stack and memory analysis, control-flow investigation, buffer-overflow research, proof-of-concept development
Security Engineering	Kernel-facing C, Linux kernel compatibility, wireless-driver engineering, GCC, Git, Python/Bash automation, technical troubleshooting

TOOLS AND TECHNOLOGIES

Assessment: Burp Suite, OWASP ZAP, Nmap/NSE, Metasploit, Nessus

Network Analysis: Wireshark, DNS tooling, TLS inspection, packet capture

Reverse Engineering / Forensics: Ghidra, Volatility, GDB

Wireless: Aircrack-ng

Programming: Python, Bash, C, SQL

Systems / Engineering: Linux, Kali Linux, Windows, GCC, Git, Linux kernel headers and build workflows

SELECTED TECHNICAL EVIDENCE

- **8 documented security findings:** authorized web assessment with evidence, root cause, impact, severity context, and remediation considerations.
- **13 driver source files:** targeted Linux wireless-driver compatibility work followed by build and real-hardware validation.
- **65+ evidence files preserved:** organized source artifacts, screenshots, scan outputs, and validation material for assessment traceability.
- **50+ PortSwigger labs completed:** hands-on web application security practice across common vulnerability classes and attack paths.

CERTIFICATIONS AND PROFESSIONAL TRAINING

Certified LLM Security Expert (CLLMSE) – Red Team Leaders	Oct 2026
Passed a specialist examination covering LLM application and AI-agent security: adversarial techniques, the OWASP Top 10 for LLM Applications (2025), AI red teaming, risk management, governance, defensive monitoring, identity and access management, secure agent architecture, lifecycle operations, and AI/LLM supply-chain risk.	
Credential verification: Verify CLLMSE credential	
Google Cybersecurity Professional Certificate – Coursera	Feb 2026
Security principles, Linux, SQL, Python, networking, security monitoring, SIEM workflows, incident response, vulnerability management, and access control.	
Tata – Cybersecurity Analyst Job Simulation, Forage	Aug 2026
IAM strategy assessment, solution design, implementation planning, and secure access workflows in a job-simulation environment.	
Introduction to Blockchain and Web3 – edX	May 2024
Blockchain, decentralized systems, Web3 architecture, and related security considerations.	
Natural Language Processing Specialization – Coursera	Apr 2023
Natural language processing and machine-learning methods.	
Java Full Stack Developer Training – CipherSchools	Feb 2023
Google Cloud Facilitator Program – Google	May 2022

EDUCATION

Bachelor of Technology (B.Tech), Computer Science and Engineering	2024
Lovely Professional University	Jalandhar, Punjab
Minor: Machine Learning	
Relevant study: Network Security, Ethical Hacking, Penetration Testing, Automation, Cloud Computing	

SECURITY METHODOLOGY

Evidence-first workflow: Discover → Validate → Investigate → Assess Impact → Recommend Remediation → Retest.
Focused on reproducible findings, responsible testing, clear technical communication, evidence preservation, and practical security improvements.